

	Email Address	Last Name	First Name	College/School Affiliation	Department Affiliation(s)	Your Title (e.g., Assistant Professor, Associate Director, Librarian, etc.)	Please enter a short synopsis of the type of research you conduct. You can list specific projects here, or give a general overview of your research.	Please list a target "end date" of your research. This should be an estimate of when a project will conclude. If you have multiple projects or your research has no	Please list any preferred skills, qualifications, or other requirements for the research.	How should students inquire about the research? (e.g., email address, link to complete a form, etc.) If the point of contact for this project is someone other than you, please list that contact name and email address.	Keywords to describe your research/lab/ongoing projects (e.g., "archival," "computational," "text analysis/interpretation," "scientific/analytic," "human subjects research," etc.)				
8/5/2026 731.11	stn@cmu.edu	Tietzmann	Nagla	Prof. Stephanie	Mellon College of Science	Physics	Full research professor	My lab studies antimicrobial peptides as a possible replacement for current antibiotics that are causing bacterial drug resistance. We take X-ray and circular dichroism data at cmu and at Cornell CDRS. We synthesize and test peptides.	Ongoing	Computer skills, some liquid handling preferred.	stn@cmu.edu	Wet lab, computational			
8/5/2026 735.50	downs@andrew.cmu.edu	Downs	Julie	Dietrich College	SOS	Professor and Head	This semester we will be working on topics around in-person data collection for human subjects research, which will involve opportunities for developing, collecting data, and analyzing a wide array of data quality tasks, helping to staff our in-person events, and general help with behavioral research tasks at every level. In addition, all new members of the lab start with general training in behavioral science methods including human subjects training certification, survey design, data visualization, and the basics of doing behavioral research. We meet weekly with the lab's PhD and undergraduate students to review progress of ongoing projects and involve lab members in a variety of tasks.	ongoing	Interest in behavioral science. Preference is given to SOS majors.	http://www.cmu.edu/dietrich/academic/psychology/index.cfm	behavioral, human subjects research				
8/5/2026 806.16	mriang@andrew.cmu.edu	Niang	Marie-Fatou	Dietrich College	Languages, Culture and Applied Linguistics	Full Professor	The student will contribute to an ongoing research and editorial project connected to the Center for Black European Studies and the Atlantic and the Journal of Black European Studies. Under close supervision, the student will conduct extensive archival and bibliographic research, identify relevant primary and secondary sources, and help create annotated bibliographies, institutional publications, and collections related to Black European Studies. Depending on the student's interests, the project may culminate in a short research monograph, annotated bibliography, or curated digital resource that will support the Journal of Black European Studies created by the Center.	'ongoing'	No previous research or editorial experience is required. The student should have an interest in political science, history, literature, cultural studies, or scholarly publishing. Reliability, intellectual curiosity, attention to detail, and a willingness to learn new research methods are the principle qualifications. Creative and programming skills welcomed.	mriang@andrew.cmu.edu	Archival research, creative, editorial research, digital humanities, bibliography and metadata, scholarly publishing, interdisciplinary humanities research, public humanities				
8/5/2026 908.58	cmcdonough@andrew.cmu.edu	McDonough	Carrie	Mellon College of Science	Chemistry	Associate Professor	Ecochemistry: We investigate how wildlife accumulate contaminants over time. Much of our recent work has focused on marine systems. It is important because these organisms are at the base of the food web and are a source of contamination to fish, other predators, and eventually humans. We evaluate different sampling devices and tools to predict tissue concentrations that would arise in organisms exposed to contaminated sediments. More recently, we're starting a project analyzing PFASs and other contaminants in bird feathers from local resident birds in collaboration with the National Aulay.	ongoing	basic chemistry knowledge, lab experience with pipetting, making solutions is desired but willing to consider anyone enthusiastic and highly motivated	contact me with statement of what they're interested in and a resume	environmental, pollution, human exposure, toxicology, ecotoxicity, toxicity, PFAS, mass spectrometry, analytical				
8/5/2026 919.05	vachani@andrew.cmu.edu	Vicky Achanti	Vicky	College of Fine Arts	Carnegie Mellon Architecture	Special Faculty, Associate Studio Professor	Analysical: We use high-resolution mass spectrometry and ion mobility to collect large amounts of data on the chemical composition of environmental and biological samples. We use and help to develop open-source software (for example, RunMass) and other tools (for example, in silico prediction of collision cross sections) to screen these big datasets to prioritize contaminants of interest and identify novel contaminants. The Urban Classroom: Sankufo Bamboo Greenhouse Synopsis: The Urban Classroom: Sankufo Bamboo Greenhouse is a community-oriented design build initiative that reimagines architectural education as a vehicle for environmental stewardship, social equity, and civic engagement. By repurposing bamboo from the former NDMAAS Spring Carnival Pavilion into a greenhouse and community classroom, the project demonstrates the potential of low-carbon, bio-based construction while advancing material reuse and circular design practices.	May 2027	Drawing, hands on building, Digital Fabrication, Data sorting and entry related to materials.	Email directly to Vicky Achanti	Circular Materiality, Design Build, Participatory Design, Bio-Based Building, Hands-on Architecture				
8/5/2026 949.56	valdes@andrew.cmu.edu	Valdes	Catarina	Dietrich College	Psychology	Special Faculty	Our lab is broadly interested in understanding how children learn from their experience. We conduct behavioral research with human participants (mostly children), meta-analyses, and computational/computer analyses. To learn more about the lab, please visit our website: https://sites.google.com/andrew.cmu.edu/csd/home	Ongoing	Prior experience with children is helpful. Students who enjoy working as part of team and gaining independence with new tasks are especially encouraged to apply.	Interested students can contact me via email and CC the lab manager, Molly Heblman (mheblman@andrew.cmu.edu). Please include a CV/resume; a cover letter is not required, but feel free to include any relevant information in the email.	Human subject research; Developmental research; Meta-analysis; Corpus analysis				
8/5/2026 1037.07	ralespin@andrew.cmu.edu	Alerjany	Nitoufar	College of Fine Arts	Architecture	Special Faculty	A central component of the project will be continuing the development of an experimental website conceived as a research instrument and cultural artifact rather than a conventional repository. The site may evolve into a digital library of spatial words, narratives, images, drawings, and artifacts that bring historical, vernacular, and speculative forms of knowledge into relation. This work may include researching culturally specific spatial concepts, developing new architectural terms and definitions, constructing thematic and epistemological connections, and testing how digital interfaces can organize knowledge through association, translation, juxtaposition, and narrative.	ongoing	Curiosity, intellectual initiative, and a willingness to explore unfamiliar ideas, methods, and materials; familiarity with Adobe Creative Suite preferred.	via email	archival, creative, artistic, speculative, interdisciplinary, experimental, editorial, critical interpretation, publication-oriented, linguistics, critical drawing investigation				
8/5/2026 1055.11	iminyeh@cmu.edu	Yeh	Iman	College of Fine Arts	School of Art	Associate Professor	I have a solo exhibition/installation at the Mattress Factory in 2027. My project involves research into the Mexican War Streets neighborhood, then the creation of small paper architectural models, sculptures of every building in the historic district (approx. 1000). Fall semester will be dedicated to research and organizing the 1000 buildings and the material approach to building. Spring semester is focused on physically building.	Fall 2027	Interest in history and architecture. Experience in architectural models and/or papermodeling. Experience in database management, organized, crafty.	iminyeh@cmu.edu	archival, sculpture, installation, diorama, architectural models and/or papermodeling, databases, craft, miniature, model, artwork				
8/5/2026 1121.31	tseltz@cmu.edu	Tersault	Lisa	Dietrich College	History	Associate Professor	Research into historic Supreme Court Cases where the Court denies voting rights. Along with research into where people decide to vote anyway.	ongoing	some knowledge of american history is helpful, but not required	tseltz@cmu.edu, email me	archival (digital), legal, constitutional, democracy, gender, race, voting				
8/5/2026 1152.51	anoyce@andrew.cmu.edu	Noyce	Abigail	Dietrich College	Neuroscience Institute	Assistant Research Professor	The Laboratory in Memory Neuroscience is led by Barbara Shinn Cunningham and Abigail Noyce. We study how humans perceive, understand, and use sensory input, especially when many things are happening at once – questions like “how do you know which sounds are important to listen to?” “How do you cope with being interrupted?” and “what kinds of information are stored in memory, and how does that change?” Research in our lab uses techniques including neuroimaging (fMRI, EEG, MEG), physiological measures (pupillometry and eye-tracking), and behavioral studies conducted both in person and online.	ongoing	We seek undergraduate research assistants who are curious about the brain and who will be kind, responsible, and reliable colleagues. Our lab uses Python, Matlab, and R, and we seek students who are comfortable in at least one of those languages. We generally seek students to commit ~10h/week during the semester to lab responsibilities.	Email Abigail Noyce with a current resume and a sentence or two about what you're hoping to learn as a member of our lab.	Neurophysiological data; Human Memory; Data visualization; Eye Tracking; Neuroscience; psychology; hearing				
8/5/2026 1255.33	calrym@andrew.cmu.edu	Smith	Calyn	School of Computer Science	Robotics	PhD Student	We are creating the software for a robot-assisted feeding demo, where a wheelchair-related robot arm will autonomously prepare food and transfer it to a user with a mobility disability. There may also be opportunities to contribute to a research project on infant learning and shared control for teaching robots to do manipulation tasks. My lab creates intelligent mobile systems for healthcare delivery at scale using wireless sensing, signal processing, and AI.	Ongoing	Experience with software development in Python (especially on teams) is highly preferred. Experience with robotics, ROS, or machine learning is preferable but not required.	Reach out to Calyn Smith at calrym@andrew.cmu.edu	robotics, software development, human subjects studies				
8/5/2026 1413.29	justin@cmu.edu	Chan	Justin	College of Engineering, School of Computer Science	ECE / SD	Assistant Professor	Mobile Health: AI-driven sensing technologies for scalable healthcare. Ubiquitous Computing Systems: Rethinking the sensing capabilities of everyday devices. Computational hearing: Real-time intelligence for superhuman hearing.	Ongoing	n/a	Email justinchan@cmu.edu	ubiquitous computing, embedded systems, wearables, mobile health, computational hearing				

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Attention-adaptive Captions in XR Live captions and translations in XR must balance two competing goals: captions should remain visible long enough to be read, but advance quickly enough to stay synchronized with ongoing conversation. Fixed timing times cannot account for differences in reading speed, caption length, or divided attention, while manually advancing every caption would interrupt the conversation. In this project, you will build an eye-tracked XR captioning prototype that adapts caption timing based on the user's visual attention. For example, a caption could remain visible while the user is still reading, advance after they stop reading, and keep the previous caption briefly recoverable. You will compare this approach with simpler alternatives, such as fixed duration, and use a variety of scenarios. Throughout the semester, you will develop the prototype, conduct a controlled user study, and investigate how attention-adaptive captions affect comprehension, synchronization, visual attention, and perceived control. Benefits Hands-on experience developing an eye-tracked adaptive XR interface Experience designing and conducting a controlled HCI study Potential to contribute to publishable research in top-tier venues such as CHI or UIST Related Work Learning Cooperative Personalized Policies from Data Data. Gehrmann et al. UIST 2019 Implicit Gaze Research for XR Systems. Sundbom et al. 2024 Predictability and Accuracy in Adaptive User Interfaces. Gajjar et al. CHI 2008 Composing Content-Aware Widgets in XR Context-aware XR systems may be able to predict which functionality a user needs, but a set of relevant functions is not yet a suitable interface. A time, niche interaction, message composer, music control, and map have very different interaction requirements. Some should remain continuously visible, some should be grouped together, and others should appear only when requested. In this project, you will build an XR prototype that transforms contextually suggested functional facets into coherent interactive bundles. You will investigate how factors such as relevance, interaction requirements, temporal demands, and relationships between functions should influence visibility, grouping, prominence, persistence, and progressive disclosure. You will compare several composition strategies, such as flat layouts, application-based grouping, build-based grouping, and evidence-based layout. Throughout the semester, you will develop the prototype, conduct a controlled user study, measure the experience of different compositional factors, and derive design guidelines for building context-aware XR interfaces. Benefits Hands-on experience designing adaptive and context-aware XR interfaces Experience conducting a controlled HCI study and analyzing mixed quantitative and qualitative data Opportunity to develop practical design guidelines for future XR systems Potential to contribute to publishable research in top-tier venues such as CHI or UIST Related Work A Benchmarking Framework for Context-aware XR Interfaces. Cho et al. UIST 2026 AMeXr: Mining Personalized Extended Reality Interfaces. Cho et al. CHI 2024 FrameXr: A tool for authoring adaptive UIs using keyframes. Wu et al. UI 2024																	
8/7/2026 11:19:5	davidindbauer@cmu.edu	Lintbauer	David	School of Computer Science	HCI	Assistant Professor		End of semester	Strong programming skills and the ability to build working prototypes Experience with WebXR (WebXR or XR Blocks), Unity or XR interface development Interest in HCI and user study research	davidindbauer@cmu.edu	HCI, Extended Reality, XR, adaptive systems						
8/7/2026 11:20:4	davidindbauer@cmu.edu	Lintbauer	David	School of Computer Science	HCI	Assistant Professor		End of semester	Strong programming skills and the ability to build working prototypes Experience with WebXR (WebXR or XR Blocks), Unity or XR interface development Interest in HCI, interaction design, and user study research	davidindbauer@cmu.edu	HCI, Extended Reality, XR, adaptive systems						
8/7/2026 11:22:0	davidindbauer@cmu.edu	Lintbauer	David	School of Computer Science	HCI	Assistant Professor		End of semester	Strong programming skills (experience in at least one of web, Unity, or Android/iOS development) Experience with gesture recognition / CV is a plus, but not required	davidindbauer@cmu.edu	HCI, Extended Reality, XR, adaptive systems						
8/7/2026 14:33:4	jimmyche@andrew.cmu.edu	Cheng	Jimmy Wei-Chun	College of Fine Arts	Architecture	Special Faculty		ongoing	Background in architecture, design, computational design, computer science, HCI, or a related field. Familiarity with architectural drawings, representations, and precedent analysis. Experience with AI tools, large language models, computer vision, or multimodal models is helpful. Experience with data organization, CSV/JSON, image datasets, or annotation workflows is helpful. Interest in UI/UX design, web prototyping, or interactive tools is a plus. Strong attention to detail and ability to work systematically with architectural data. Ability to document research processes clearly and work independently. For students interested in the user study component, experience with qualitative research, surveys, interviews, or data analysis is beneficial but not required. Preferred qualifications include: Background in architecture, design, computational design, computer science, HCI, or a related field. Familiarity with architectural drawings, representations, and precedent analysis. Experience with AI tools, large language models, computer vision, or multimodal models is helpful. Experience with data organization, CSV/JSON, image datasets, or annotation workflows is helpful. Interest in UI/UX design, web prototyping, or interactive tools is a plus. Strong attention to detail and ability to work systematically with architectural data. Ability to document research processes clearly and work independently. For students interested in the user study component, experience with qualitative research, surveys, interviews, or data analysis is beneficial but not required.	email to jimmyche@andrew.cmu.edu	Computational design, artificial intelligence, multimodal AI, architectural representation, information retrieval, human-AI interaction, interface design, design cognition, dataset development, prototyping						
8/7/2026 20:47:2	qiaosin@andrew.cmu.edu	Wang	Qiao (Chelsea)	School of Computer Science	Human-Computer Interaction Institute (HCII)	Postdoc Fellow		end of semester, can be renewed	Qualitative data analysis (coding), literature review, user interviews and/or workshop. Ability to understand and synthesize cognitive science and psychological theories. Skills and experience in NLP and annotating conversation data would be a plus.	Please email me with your resume/CV 1-2 sentences about why you are interested in this project and what you are trying to get out of it. If you have relevant research experience and skills, please also highlight them in your email.	human subjects research, theory and framework, qualitative data analysis						
8/9/2026 13:58:5	vnoel@andrew.cmu.edu	Noel	Vernelle	College of Fine Arts	Architecture	Associate Professor		Ongoing	Skills required will depend on the research activities the student engages in.	https://forms.gle/8UjR8pDz8UW6d4	computational, design, making, fabrication, workflow, software, digital modeling.						

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							Adaptive Reasoning for Long-Horizon Manipulation Description: How can robots solve long-horizon manipulation tasks in previously unseen environments? This research explores adaptive reasoning that combines prior knowledge from foundation models with online interaction to enable robots to reason about the physical world, plan, and adapt. Potential research directions include world model-guided self-correction of LLM-generated plans, active information gathering to reveal hidden object properties, online adaptation to geometric and contact constraints, and scalable learning of general-purpose world models or reasoning agents for robotic planning. Our long-term vision is to develop robots that leverage internet-scale prior knowledge while continually improving through interaction with the physical world. Link(s) to Relevant Papers: Reflective Planning: Vision-Language Models for Multi-Stage Long-Horizon Robotic Manipulation RobotWits: Unexpected Challenges for Robotic Creative Problem Solving KIDR: A Physical Reasoning Benchmark for Robot Learning and Planning MILench: A Benchmark for Intuitive Robotic Manipulation Contact: Amber Li -amberli2@gandrew.cmu.edu										
8/9/2026 16:01:41	dhw@gandrew.cmu.edu	Held	David	School of Computer Science	Robotics	Associate Professor	Adaptive Policies for Cross-Embodiment Devisious Manipulation Description: How can we build foundation models for devisious manipulation that generalize across different robot hands, tasks, and environments? This research explores learning algorithms for devisious multi-fingered manipulation by combining large-scale pretraining, meta-learning, and fast-time adaptation. Potential research directions include cross-embodiment learning, online adaptation to novel objects and environments, scalable learning from large-scale human and robot interaction data, learning adaptive latent representations that capture embodiment-specific properties through interaction, and incorporating contact-aware policy learning with force or tactile feedback for robust interaction in contact-rich tasks. Our goal is to enable effective transfer across diverse robotic hands and challenging manipulation tasks, including functional grasping, in-hand manipulation, and tool use. Link(s) to Relevant Papers: [2410.02478] Cross-Embodiment Devisious Grasping with Reinforcement Learning [2402.08278] DeafFormer: Cross-Embodied Devisious Manipulation via History-Conditioned Transformer [2402.00918] LmMetaGrasp: Diffusion Model with Morphology Awareness for Cross-Embodiment Devisious Grasp Generation [2410.07702] SymMetaGrasp: Group-A Unified Representation of Robot and Object Interaction for Cross-Embodiment Devisious Grasping Contact: Ying Yuan -yinyuan@gandrew.cmu.edu	Ongoing	None	Contact the students listed above associated with the above projects.	robotics, reinforcement learning, 3D vision						
							Virtual reality can show you an object, but making it feel solid is much harder. One trick is to place a real surface exactly where the virtual object appears, so that touching the illusion produces a real contact force. It works because your brain predicts what your hand will feel before the hand arrives. In our first study we broke that prediction completely on about one reach in five, the physical surface was quietly removed and the hand passed straight through something that looked solid. In 12 people, electroencephalography (EEG) showed a large brain response to the touch that never came. Similar responses have been reported for missing sounds [8] and for mismatched touch in virtual reality [1, 2]. That was the easy case. Real interfaces fall in subtler ways – the object looks like polished metal and feels like foam. This project asks where the limit is: how different does the felt surface have to be from the seen surface before the brain notices, and does the response grow smoothly with the mismatch or switch on at a threshold? The student will help build a rig that presents surfaces of varying texture behind the same rendered object, then measure how the neural responses scale. The answer sets how good the haptics in a surgical trainer or a prosthetic hand actually need to be. Here the participant has no say in when the mismatch occurs – a deliberate contrast with our parallel driving project. Benefits Experience designing an experiment in which the measurement chain is engineered and verified rather than assumed. Potential to contribute to publishable research in top-tier venues Related Work Detecting Visual-Haptic Mismatches in Virtual Reality using the Prediction Error Neutrality of Event-Related Brain Potentials, Gehke, Aiman, Lopes, Chen, Singh, Chen, Lin, and Diamant, CHI 2019. Neural sources of prediction errors detect unrealistic VR interactions, Gehke, Lopes, Rug, Aiman, and Diamant, Journal of Neural Engineering 2022. Hearing Illusions: Human Auditory Processing Relies on Preactivation of Sound-Specific Brain Activity Patterns, SanMiguel, Wildmann, Bendure, Trujillo-Barreto, and Schröger, Journal of Neuroscience 2018.	End of semester	Unity development in C#, including hand tracking and collision event handling Ability to build and debug simple electromechanical hardware (potentiometer or servo actuation, microcontroller control, sensor logging)	david@bauer@cmu.edu	Virtual reality, biosignals, EEG, adaptive interfaces						
8/10/2026 13:52:14	david@bauer@cmu.edu	Lin/Bauer	David	School of Computer Science	HCB	Assistant Professor	Who is Steering: Neural Responses to Covert Automation Intervention During Manual Driving Can increasingly steer with you rather than for you. Lane-keeping systems push on the same wheel you are already holding, which blurs the line between your action and the car's. People feel less in control as automation increases [1, 2], and brain responses differ depending on whether an error is blamed on yourself or on something outside you [8]. In our 15-person driving simulator study, the wheel was covertly nudged on a few of curves and participants were never told. Electroencephalography (EEG) showed a clear response about a quarter second later. The study has fixable flaws. The nudge was always the same size, and it shifted the visual scene as well as the wheel, so we cannot say what the driver was actually reacting to. This project rebuilds the simulator so the nudge ranges from unnoticeable to obvious, and asks the driver after each one how much of the motion felt like their own doing. Unlike the virtual reality project, the participants here have control over the violation. Benefits Experience identifying confounds in an existing dataset and then designing the study that removes them. Potential to contribute to publishable research in top-tier venues Related Work Automation Technology and Sense of Control: A Window on Human Agency, Beberian, Saman, Le Blay, and Haggard, PLoS ONE 2012. Sense of agency in the human brain, Haggard, Nature Reviews Neuroscience 2017. Violating body movement semantics: Neural signatures of self-generated and external-generated events, Padoa, Conzalez-Franco, Sanchez-Vives, Stiles, and Rodriguez-Fornells, NeuroImage 2016. We have a number of exploratory projects on developing AI techniques for biomedical image analysis and autonomous experimentation. More information about our research can be found on our lab website. http://ics.cmu.edu/~mxi/	End of semester	Unity development in C#, including force-feedback wheel integration and vehicle dynamics Ability to build and validate hardware event timing (microcontroller trigger generation, photodiode or oscilloscope verification of latency)	david@bauer@cmu.edu	Driving, Simulator, biosignals, EEG, adaptive interfaces, HCI						
8/10/2026 17:54:12	mxi@gandrew.cmu.edu	Xu	Min	School of Computer Science	Computational Biology	Associate Professor	The following are our recent publications and software projects as examples https://scholar.google.com/citations?hl=en&user=Y5CpGAAA&view_op=list_works&sortby=update https://github.com/rutabati/atom	ongoing	Python programming	mxi@gandrew.cmu.edu	Biomedical image analysis, Computer Vision, Machine Learning, Autonomous Experimentation, Robotics						
							Software Prototyping + Human Subject Research (Interviews + Software task Analysis) Motivation What happens when AI agents can directly use computers on your behalf – and how do we make that safe? This project dives into the emerging space of computer use agents with a dual focus: building powerful, real-world security assistants and designing the safeguards that keep them trustworthy. On one side, we explore how agents can actively help users manage security across everyday workflows – monitoring activity, responding to threats, automating protections, and navigating complex systems with minimal human intervention. On the other, we tackle a harder problem: securing the agents themselves. How do you constrain an agent's actions, verify its decisions, and prevent unsafe or unintended behavior when it has real access to systems? This is a hands-on, systems-oriented project for people interested in building agents and conducting human subject studies to evaluate the prototypes. We are looking for research assistants to help with a project examining decision-making in the U.S. Supreme Court. We are interested in understanding how characteristics of justices and their interactions during oral arguments relate to judicial decision-making. We are looking for a couple of students to assist with evaluating Supreme Court oral argument transcripts and analyzing these transcripts using large language models (LLMs). There may also be opportunities to contribute to statistical analyses. This project is a great fit for students interested in psychology, medical sciences, computational social science, or the use of AI and natural language processing in research.	Ongoing	- Experience building or experimenting with AI agents (e.g., tool use, autonomous workflows, LLM integrations) - Strong programming skills (Python preferred) - Familiarity with APIs, automation frameworks, or system scripting - Interest in cybersecurity, system safety, or adversarial thinking - Understanding of operating systems or system-level behavior - Curiosity about safe AI, agent alignment, and real-world deployment challenges - Bonus: experience with sandboxing, permissions systems, or security tooling	kmonte@gandrew.cmu.edu	Human Computer Interaction						
8/10/2026 22:05:18	kmonte@gandrew.cmu.edu	Montano	Kyril	School of Computer Science	HCB	PHD Student											
8/13/2026 14:12:12	kkeedrick@gandrew.cmu.edu	Keedrick	Kara	Dietrich College	Institute for Complex Social Dynamics	Postdoctoral Fellow		ongoing	interests in political science, decision making or psychology	kkeedrick@gandrew.cmu.edu	Computational, archival, text analysis, NLP, large language models (LLMs), behavioral science, quantitative analysis						

Timestamp	Email Address	Last Name	First Name	College/School Affiliation	Department Affiliation(s)	Your Title (e.g., Assistant Professor, Associate Director, Librarian, etc.)	Please enter a short synopsis of the type of research you conduct. You can list specific projects here, or give a general overview of your research.	Please list a target "end date" of your research. This should be an estimate of when a project will conclude. If you have multiple projects or your research has no	Please list any preferred skills, qualifications, or other requirements for the research.	How should students inquire about the research? (e.g., email address, link to complete a form, etc.) If the point of contact for this project is someone other than you, please list a contact name and email address.	Keywords to describe your research/lab/ongoing project(s) (e.g., "technical," "computational," "text analysis/analysis," "scientific/arts/cultural," "human subjects research," etc.)									
							ChangLing Lab (led by David Mortensen in the Language Technologies Institute) is currently seeking one or more undergraduate research assistants in the following areas: LLM Agents for Comparative Linguistic Reconstruction Starting in the 19th century, philologists and linguists have developed a method for reconstructing the pronunciation of the words in an ancestral language given the pronunciation of the corresponding words in its modern descendants (the Comparative Method). The models of linguistic reconstruction and changes produced by the method are essentially computer programs, and our lab has made considerable progress treating parts of the Comparative Method as a computational problem. During this summer, we plan to implement the whole comparative method (which we have formalized as a discrete optimization problem) using LLM agents. This role will involve both linguistics (especially historical phonology) and machine learning (agentic AI). LLM Agents for Discrete Optimization Problems LLMs have shown considerable promise in finding approximate solutions to reasoning problems that appear to be intractable. Our work in this area, for example, that symbolic solvers for sound law induction (part of the Comparative Method) and more general systems of logical rules can be distilled from the traces of LLM agents (https://arxiv.org/abs/2603.05485). We are seeking one or more undergraduate research assistants to help us to extend this work, both through the introduction of new techniques and the application of existing techniques to new classes of problems. These students will work closely with the students who are working on comparative linguistic reconstruction. LLMs and Lexical Change Language is a dynamic system that changes every time it is used and thus varies in space, time, and social context. Large Language Models, in contrast, are typically trained cross-sectionally, as if time did not exist, and have difficulty reasoning about time, not just across the history of a language but even in a single piece of discourse. We are seeking one or two students to work on change in the meaning of terms over time as it relates to LLMs. Specifically, we seek to understand what knowledge LLMs have of historical change in the meaning and use of words, what capacity they have to reason about differences produced by this change, and how this ability can be enhanced.	ongoing	- Strong coding skills in Python - At least one course in linguistics - At least one course with a strong machine learning focus - Experience with one or more widely-used machine learning frameworks (PyTorch, Huggingface TGI, vLLM, etc.) - Strong interest in general issues of language, linguistics, and logical reasoning	Applicants should complete the form at https://docs.google.com/forms/d/e/1FAIpQL5o7YV9VRLsc4K3BZ7sYxJJXVhK3-dyGdWuT9-TCn5BBA/viewform?usp=publish-editor	language change, agentic AI, machine learning, historical semantics,									
8/10/2026 6:59:3	dromonagis@cmu.edu	Mortensen	David	School of Computer Science	Language Technologies Institute	Associate Research Professor	We are currently developing next-generation Vision-Language-Action (VLA) models and deploying them onto advanced humanoid robots. Our lab develops ultra-efficient fine-tuning techniques that empower these VLA models to master highly complex, real-world tasks. This deployment phase is conducted in close collaboration with Fujitsu Research as part of their pioneering Physical AI initiative.	Ongoing	- VLA & Multimodal AI: Vision-Language-Action frameworks, VLAs, model fine-tuning - PET & Compression: LLMs, model optimization - Embedded AI: imitation learning, reinforcement learning (RL), optimal control - Humanoid Robotics, Edge Deployment	Please apply via this form: https://docs.google.com/forms/d/e/1FAIpQL5o-gf78YhNA5AAUf9Rl_WuTPU3UonFA_Hm9hLj_G5Bh-pxe-wae-form?usp=sharing&ouid=1058864432212134179	Control, ML, Vision-Language-Action (VLA) models, Physical AI, humanoid robotics, Language-guided control									
8/10/2026 10:22:	tommy@andrew.cmu.edu	Yang	Tommy	College of Fine Arts	Architecture	Special Faculty	In matter / floating matter is a multidisciplinary design research lab that explores how stories, materials, and territories shape the built environment. We ground our work on many occasions to vernacular design in response to socio-ecological agencies. Framed through the lenses of Traditional Ecological Knowledge (TEK) and storytelling, our work explores how spatial practices, vernacular and materiality can reveal alternative worlds. The projects range from socio-ecological analysis of east and southeast asia to prototyping hybridized workflows that integrate TEK to advanced fabrication techniques, to ethnographies among digital platforms like games. As part of the lab, you will have opportunities to build and present refined portfolio works of systemic confrontation, asking what it means to design with practices' states' that have historically read as evasion, refusal, or resistance to be generated.	ongoing	There are no prerequisites for research. One expectation is that students interested come with an open mind and are willing to experiment. Depending on the research interest, you will be trained in ethnographies, illustrations, film, the digital fabrication lab, and wood shop upon joining the design family.	Interested students in any of the research mentioned above, please send a cover letter as well as a resume to my preferred email: tommy@andrew.cmu.edu , describing your interests	design, ecology, vernacular, traditional ecological knowledge, fabrication, digital modeling, storytelling									
8/21/2026 8:51:4	veebster@andrew.cmu.edu	Webster-Wood	Vickie	College of Engineering	Mechanical Engineering	Associate Professor	Looking for an undergrad to troubleshoot and improve upon an existing hardware system for use in a wet lab environment. The system includes multiple syringe pumps controlled purely through an Arduino interface to enable parallel hydraulic actuation of multi-actuator aquatic soft robots. Students interested should have relevant skills including 3D printing, Arduino coding, electrical circuit design, and general mechanics prototyping experience.	12/2026 (then future projects become available)	3D printing, Arduino coding, electrical circuit design, and general mechanics prototyping experience	email a resume and brief statement of interest to Avery Williamson (avery@andrew.cmu.edu) and cc (veebster@andrew.cmu.edu). Include the tag ECE Spring Pump in the subject line.	mechatronics, systems integration									
9/8/2026 11:50:9	bockias@jgill.edu	Boockia	Irina	University of Pittsburgh	Pathology	Associate Professor			Command line, Python, R	email bockias@jgill.edu	Computational biology, genomics, Epigenetics									
9/10/2026 15:07:	josear@andrew.cmu.edu	Andriano Martorel	Jose	Dietrich College	Social and Decision Sciences	PhD Student	I study how people infer expertise from observable cues in online health communication. Tl	December 2026	Advanced experience creating highly realistic AI-generated content. In general, an interest in working across historical data/science	Please email Jose Andriano Martorel at josear@andrew.cmu.edu	experimental									
9/11/2026 11:16:	kmcdonough@andrew.cmu.edu	McDonough	Katherine	Dietrich College	History	Assistant Professor of History	I am a historian who develops new computational methodologies for analyzing historical data	ongoing	Basic (any of these are useful): python, pytorch/ML framework. More advanced (again, any assortment, not all in combination	Email Professor Katherine McDonough: kmcdonough@andrew.cmu.edu	computational history, spatial history, history									
8/5/2026 9:49:56	bockstaller@cmu.edu	Bockstaller	Michael	College of Engineering	Department of Materials Science and Engineering	Professor	We are developing a new class of polymer-based hybrid materials that feature novel microstructures which give rise to unique property combinations, such as self-healing or shape-memory behavior. These hybrid materials are realized via the bottom-up assembly of polymer-grafted nanoparticle building blocks. Our goal is to elucidate structure-property-processing-performance relations in this new class of materials. To accomplish this goal, we utilize a wide range of physical characterization techniques, including electron imaging, X-ray scattering, as well as mechanical characterization.	ongoing	Interest in polymer materials and working in teams	Email: bockstaller@cmu.edu	wet lab/experimental									